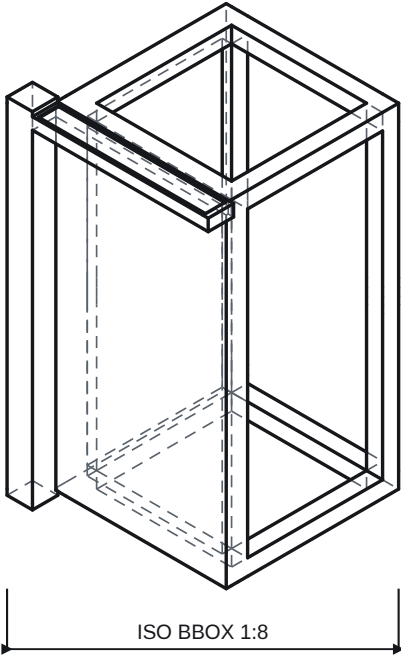


READABLE THEORY OF THE MACHINE

HLR of PR 34 assembly STEP. Interior remains plastic, screened, animal-safe.

ISOMETRIC ASSEMBLY

SCALE 1:8 | OCCT HLR
DRAFT-MEASURED fe8f875a80b3 PR 34



SYSTEM THEORY

250 mm block perimeter; 500 mm first tower.
P01-P08: protected continuous VIN/GND/I2C/UID/P08.
P09-P12: indexed GMSL2 only while stationary.
Raw MIPI stays binder-local and Tachyon-local.
Pinch is break-before-move; video off while rolling.
M1/Tachyon brick never enters the animal cage.

LOCKED SAFETY BOUNDARY

Upper-third molt keep-out 142 mm CALCULATED.
Ceiling/vents: <= 0.8 mm non-metal LOCK.
B20 is the continuous wet-side barrier (PR 34 STEP).
Door front/side; nymph-visible gap <=0.50 TARGET.
No taxon inferred: no live-mantis photo supplied.

MEASURED THIS RUN (DRAFT CAD/EE)

CAD-01/#28 fe8f875a80b3 PR 34 STEP (read-only)
CAD-02/#29 cdd523c55a63 OCCT carriage STEP
CAD-02 does not admit PR 34 as released parent.
EE-23..26 PRs 37-40 (read-only)
No Particle camera SKU or CSI pin table drawn.

DRAWING STATUS

Projected views: OCCT HLRBRep of STEP solids.
S08 is a review diagram; KiCad is circuit authority.
S09 uses datasheet envelopes, not a brick STEP.
Not #31 shop-release. Not first-article.

0 1632486480
mm @ 1:8

COLOR KEY (SHOP LINE ART)

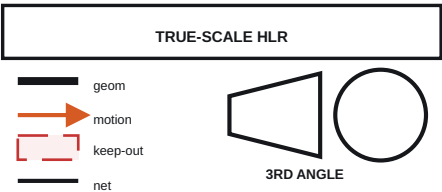
geometry motion keep-out electrical net

PARTICLE-BASE MANTIS TERRARIUM

COVER / THEORY

DRAFT CAD - NOT SHOP-RELEASE (#31) / NOT FIRST ARTICLE

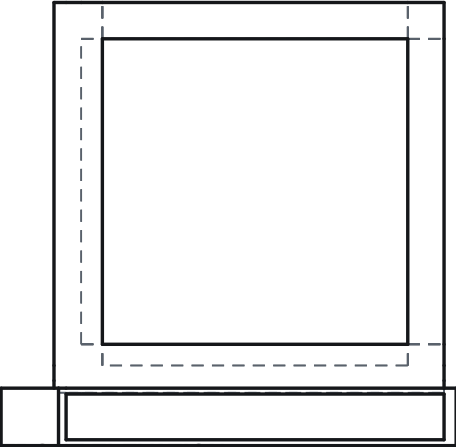
Units: mm | Projection: THIRD-ANGLE (ASME Y14.3) | Color: nets / motion / keep-out only



SCALE 1:8	DATE 2026-08-20	REV B-41P
SHEET 01 OF 12 / S00	STATUS DRAFT CAD	DRAWN GBG#41
DWG MT-S00	FORMAT A3 SVG/PDF	TOL AS NOTED

TOP PLAN

SCALE 1:5 | OCCT HLR
DRAFT-MEASURED PR 34



FRONT ELEVATION

SCALE 1:5 | OCCT HLR

RIGHT ELEVATION

SCALE 1:5 | OCCT HLR

B20 BARRIER

SCALE 1:5 | OCCT HLR
KEEP-OUT

LOCKED ENVELOPE

Exterior: 250 W x 250 D x 500 H LOCK.
Module datum repeats at 250 mm.
Nominal clear: 202 x 202 x 427 CALCULATED.
HLR bbox X -43..258,
Y -43..250, Z -8..508.

PROJECTION

Third-angle from PR 34 assembly STEP.
Front is the door/view cassette face (Y=0).
Rail offset 5 mm REF outside frame.
Hidden lines: OCCT HLRBRep. CAD-02 not parent.

INTERFACE DATUMS

A: exterior block face (frame.world origin).
B: cassette seat plane.
C: rail conductor centerline (external).
D: false-bottom tray 25 mm REF.
Molt datum: 142 mm CALCULATED.

516 CAD-01 BBOX

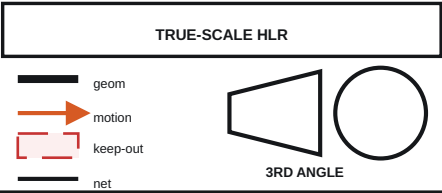
301 CAD-01 BBOX

0 1020304050
mm @ 1:5

PARTICLE-BASE MANTIS TERRARIUM
ORTHOGRAPHIC ASSEMBLY

DRAFT CAD - NOT SHOP-RELEASE (#31) / NOT FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3) | Color: nets / motion / keep-out only



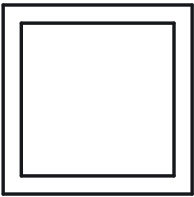
SCALE 1:5	DATE 2026-08-20	REV B-41P
SHEET 02 OF 12 / S01	STATUS DRAFT CAD	DRAWN GBG#41
DWG MT-S01	FORMAT A3 SVG/PDF	TOL AS NOTED

UNIQUE-PART HLR ALONG SERVICE AXIS

SCALE AS NOTED | OCCT HLR
CAD-01 unique STEP + CAD-02 unique STEP. Not a fused explode of one assembly.

1:10

B10



1:5

B02



1:12

B05



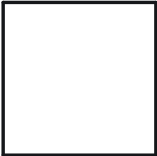
1:1

B03



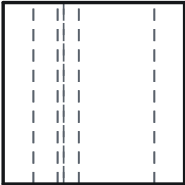
1:10

B14



1:1

B01



CEILING FRAME

EDGE 250

VIEW CASSETTE

SPLICE

FALSE BOTTOM

CORNER

RAIL / CARRIAGE / BINDER GROUP

SCALE AS NOTED | OCCT HLR



B18



B22



B29



B50



B42

BRICK IS NOT A STEP THIS RUN

EXPLOSION RULE

Each balloon is an HLR of that unique-part STEP, not a generate.py box.
CAD-01 unique: B01-B04, B10, B18/B51/B52, B05-B07, B12-B14, B20.
CAD-02 unique: B22-B27 pinch, B28/B29/B50 binder. No animal-side metal.

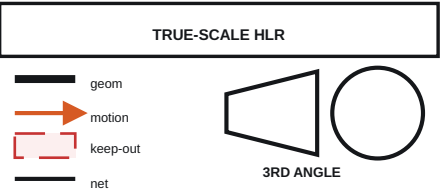
INSTALL ORDER

1 perimeter -> 2 faces -> 3 ceiling -> 4 false bottom.
5 animal-isolated rail -> 6 carriage -> 7 SerDes binder -> 8 brick hub.
First article requires leak, pinch, pull-off and nymph-gap checks.

PARTICLE-BASE MANTIS TERRARIUM
EXPLODED ASSEMBLY

DRAFT CAD - NOT SHOP-RELEASE (#31) / NOT FIRST ARTICLE

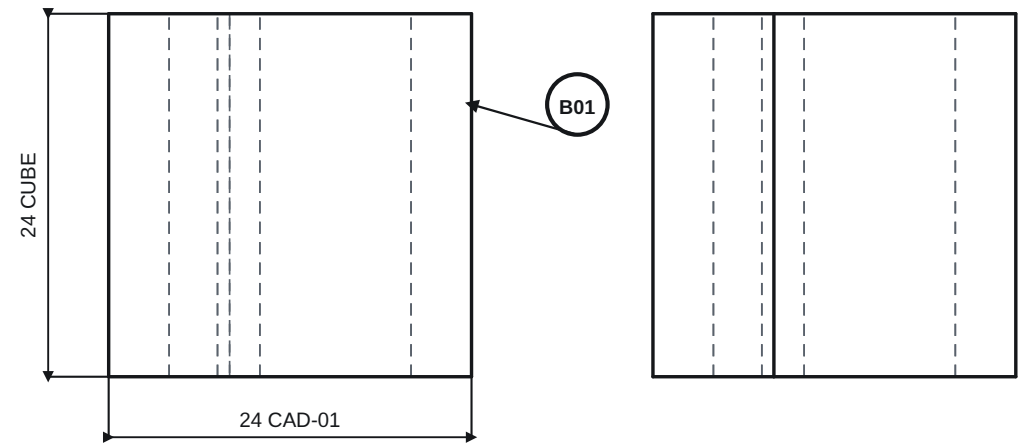
Units: mm | Projection: THIRD-ANGLE (ASME Y14.3) | Color: nets / motion / keep-out only



SCALE AS NOTED	DATE 2026-08-20	REV B-41P
SHEET 03 OF 12 / S02	STATUS DRAFT CAD	DRAWN GBG#41
DWG MT-S02	FORMAT A3 SVG/PDF	TOL AS NOTED

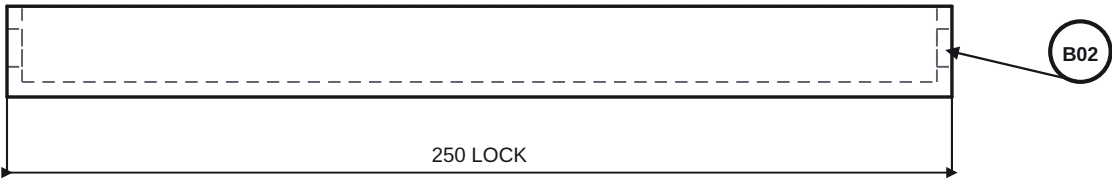
CORNER BLOCK B01

SCALE 2:1 | OCCT HLR
DRAFT-MEASURED PR 34



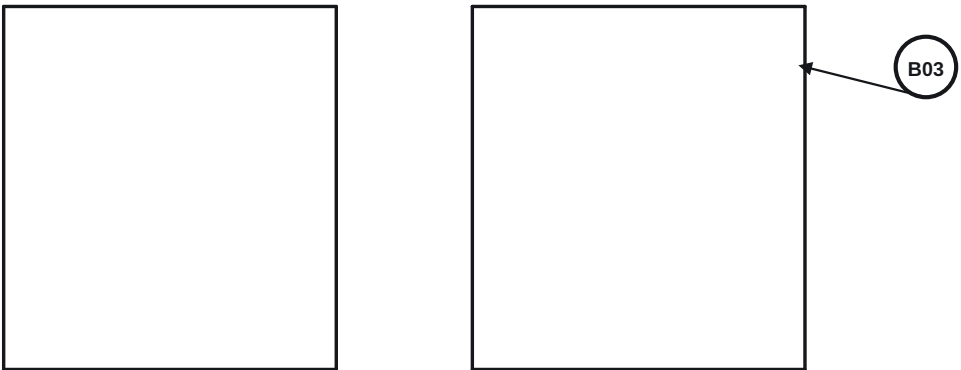
EDGE BLOCK B02

SCALE 1:2 | OCCT HLR



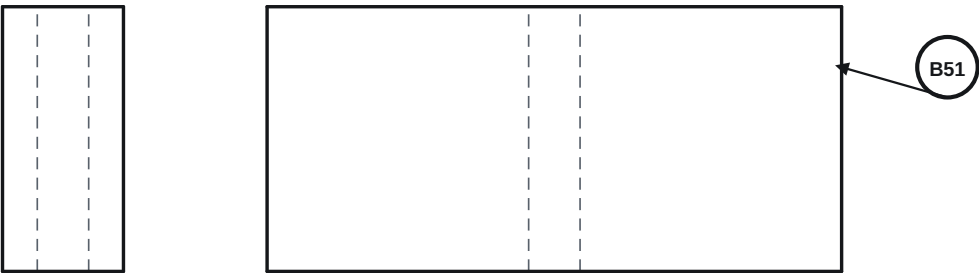
MID-SPAN SPLICE B03

SCALE 2:1 | OCCT HLR



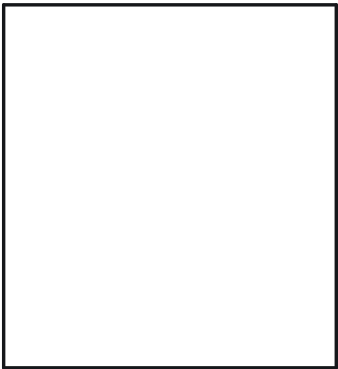
ROUTE END STOP B51

SCALE 2:1 | OCCT HLR



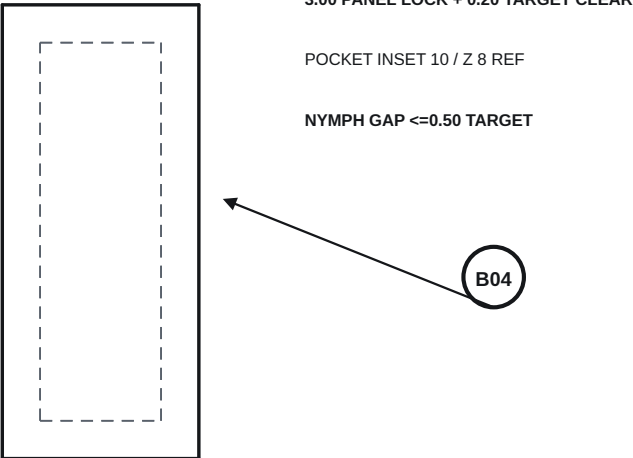
DETAIL A - B03 SPLICE HLR

SCALE 2:1 | OCCT HLR



DETAIL B - B04 RETAINER (RIGHT)

SCALE 5:1 | OCCT HLR



SHEAR BY TONGUE; B21 IS NOT A CORNER ROLL SOLID

CAD-01 MEASURED UNIQUE PARTS

B01 24 cube; B02 250 x 24 x 24; B03 22 x 24 x 24. Kerf 0.15 REF.
Views are HLR of PR 34 unique-part STEP. Not shop-release.

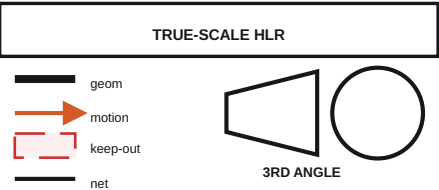
LOAD PATH / ROUTES

Panel -> gasket -> pocket wall -> perimeter block.
B21: top-front and vertical routes are independent; no corner roll.
B51 8 x 38 x 17.5; bore 3.4 UNVERIFIED PN.

PARTICLE-BASE MANTIS TERRARIUM
PERIMETER BLOCKS

DRAFT CAD - NOT SHOP-RELEASE (#31) / NOT FIRST ARTICLE

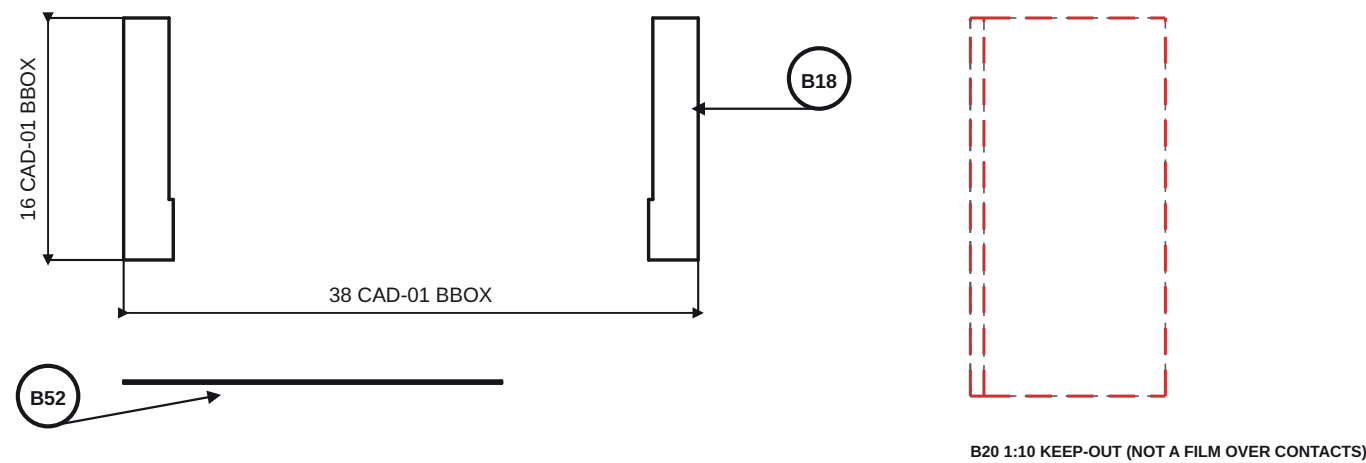
Units: mm | Projection: THIRD-ANGLE (ASME Y14.3) | Color: nets / motion / keep-out only



SCALE 2:1 / 1:2	DATE 2026-08-20	REV B-41P
SHEET 04 OF 12 / S03	STATUS DRAFT CAD	DRAWN GBG#41
DWG MT-S03	FORMAT A3 SVG/PDF	TOL AS NOTED

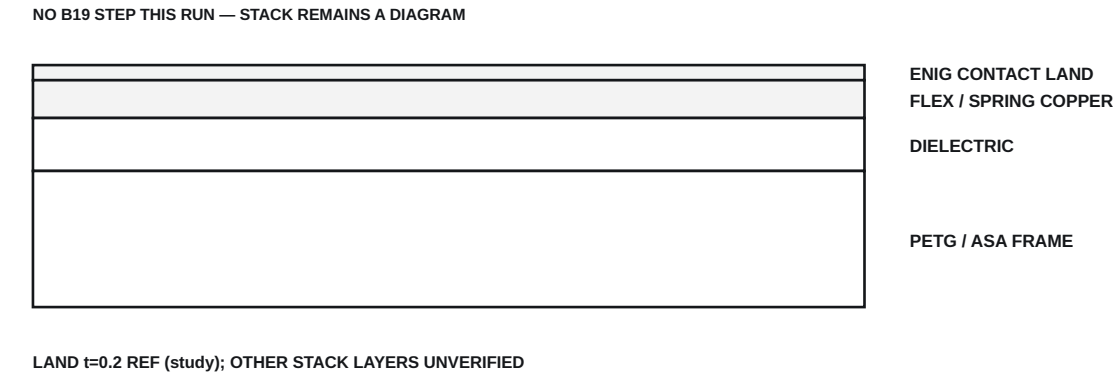
RAIL CHANNEL SECTION C-C

SCALE 2:1 | OCCT HLR
B18-rail-channel-250.step cut at X=125, HLR along +X



DETAIL D - STRIP STACK-UP

SCALE NTS - DO NOT SCALE

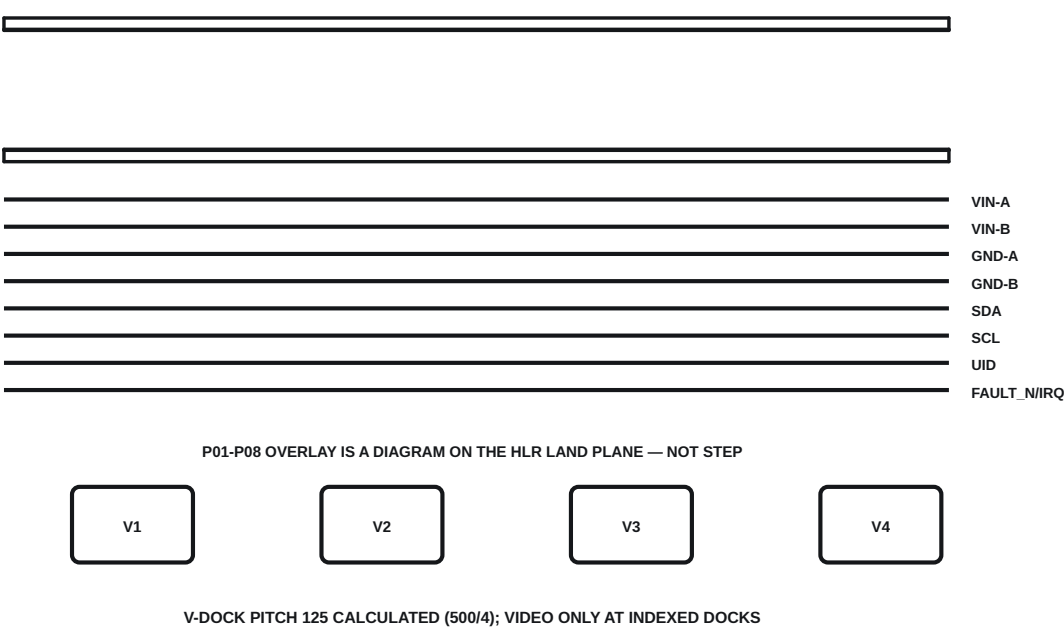


CAD-01 RAIL MEASURE

B18 250: 250 x 38 x 16; 500 span sibling.
Offset 5 REF; B52 lip 1.5; slot 0.4 REF.
Section is a true half-space cut of the B18 STEP, then HLR.
B20 is the wet-side barrier; it is not a film over contacts.

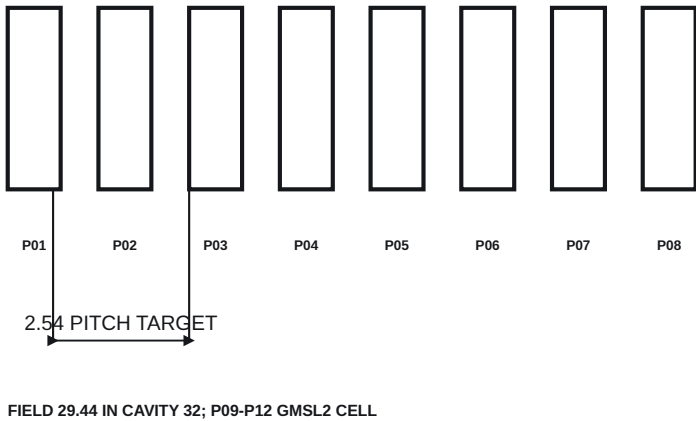
LONGITUDINAL B18 250 TOP

SCALE 1:2 | OCCT HLR



DETAIL E - CONTACT LANDS

SCALE NTS - DO NOT SCALE



FAILURE CONTROL

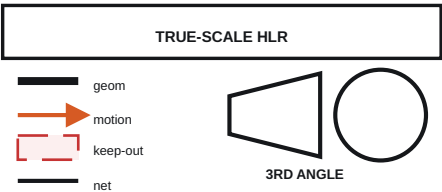
S1 first travel opens Q1 before B27 lift (CAD-02 q=1.0 TARGET).
Bent HSD pogo -> link loss; no fallback to raw MIPI on the strip.
Each V-dock is point-to-point to one MAX96724 input.
B52 is a labyrinth wiper, not a hermetic cover. EE-23 series UNVERIFIED.

PARTICLE-BASE MANTIS TERRARIUM

RAIL + STRIP

DRAFT CAD - NOT SHOP-RELEASE (#31) / NOT FIRST ARTICLE

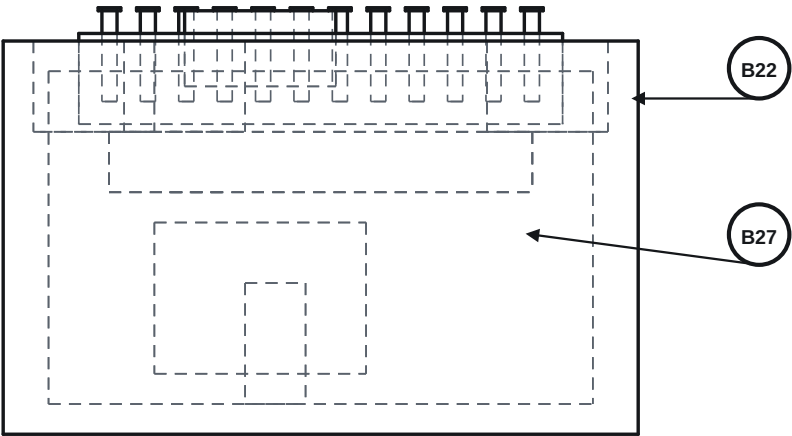
Units: mm | Projection: THIRD-ANGLE (ASME Y14.3) | Color: nets / motion / keep-out only



SCALE 2:1 / 1:2	DATE 2026-08-20	REV B-41P
SHEET 05 OF 12 / S04	STATUS DRAFT CAD	DRAWN GBG#41
DWG MT-S04	FORMAT A3 SVG/PDF	TOL AS NOTED

LOCKED STATE Q=0

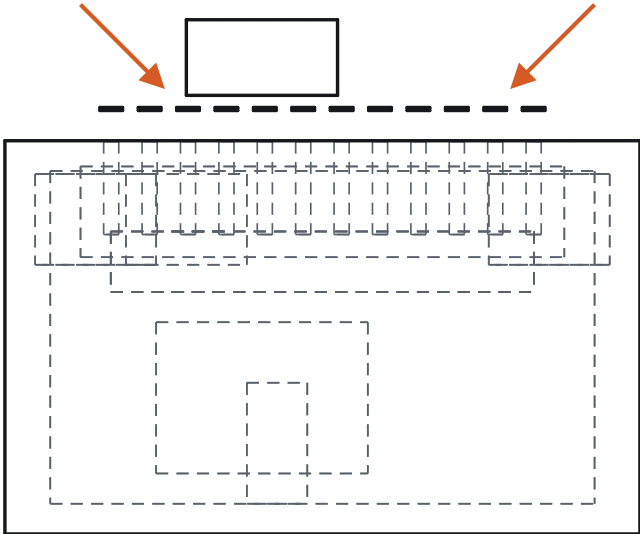
SCALE 2:1 | OCCT HLR
CAD-02 OCCT posed assembly. PR 34 not admitted as parent.



LOCKED q=0 B22 60x42x28 LIFT PAWL IN

PINCHED / TRAVEL Q=5

SCALE 2:1 | OCCT HLR



q=5 ROLLING; LIFT 2.2 THROW / 1.8 CLEAR

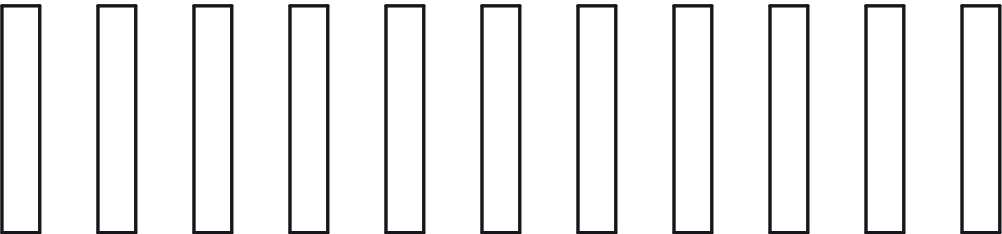
CAD-02 PINCH Q (MM)

SCALE NTS - DO NOT SCALE

q=0 LOCKED S1 closed; pawl in
q=1 S1_OPEN Q1 off; contacts seated
q=2.2 PINCH_SAFE lift pawl retracts
q=4 CLEAR lift=2.2; still locked
q=5 ROLLING translation pawl out

DETAIL F - B27 ARRAY + B25 POCKET

SCALE 5:1 | OCCT HLR



B25 SPRING PN UNVERIFIED (POCKET ONLY)

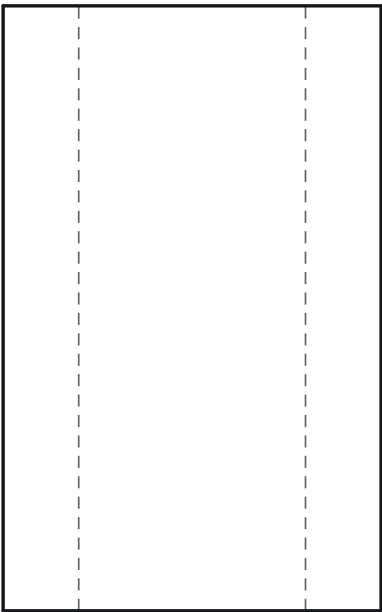
PINCH FORCE 15-25 N TARGET

B27 PROXY d=1 / BARREL 6 UNVERIFIED

MIPI IS NOT ON THESE CONTACTS

ADR-002: HARDWARE PAWLS, NOT A TIMED DELAY

0.60 WORK COMPRESSION TARGET

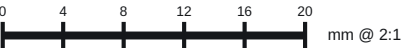


MECHANISM NOTE - PINCH-LIFT / ROLL

States: LOCKED / S1_OPEN / PINCH_SAFE / CONTACTS_CLEAR / ROLLING.
What moves: B24/B27 lift then rollers; rail/B19 stays grounded.
q=1 opens S1 envelope; contacts stay seated until q=2.2.
Lift pawl blocks Z until PINCH_SAFE. Throw 2.2 covers 1.8 clearance.
Translation pawl blocks roll until lift >= clearance. Speed cannot skip a pawl.
EE-24 S1/S2/Q1 parts UNVERIFIED; electrical dwell is a later independent gate.
Chosen over screws for one-hand relocate without opening the cage.

FAILURE / ACCEPTANCE

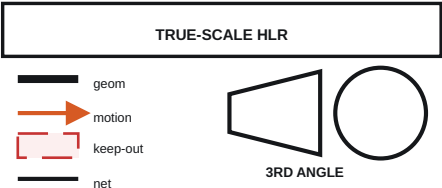
Interrupted pinch: keep Q1 off; require a clean full remate.
Bent HSD: training timeout; Q1 latches off. No MIPI fallback.
B51 retains loaded carriage (EE-26 route-retention UNVERIFIED).
No conductor may breach B20.



PARTICLE-BASE MANTIS TERRARIUM
CARRIAGE MECHANISM

DRAFT CAD - NOT SHOP-RELEASE (#31) / NOT FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3) | Color: nets / motion / keep-out only



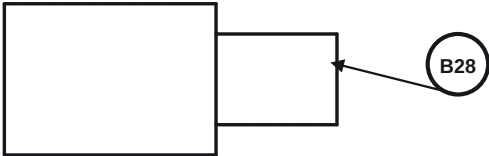
SCALE 2:1	DATE 2026-08-20	REV B-41P
SHEET 06 OF 12 / S05	STATUS DRAFT CAD	DRAWN GBG#41
DWG MT-S05	FORMAT A3 SVG/PDF	TOL AS NOTED

UNIVERSAL SHOE B28

SCALE 2:1 | OCCT HLR

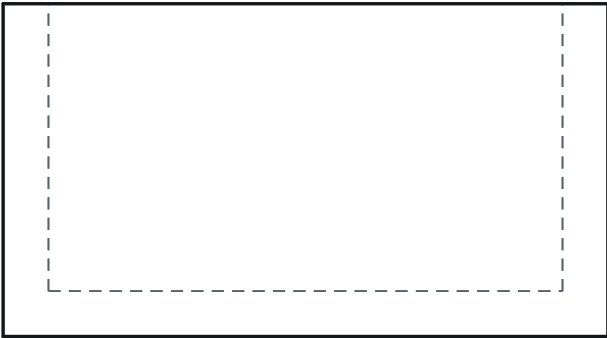


B28 SHOE 24x14x10 CAD-02

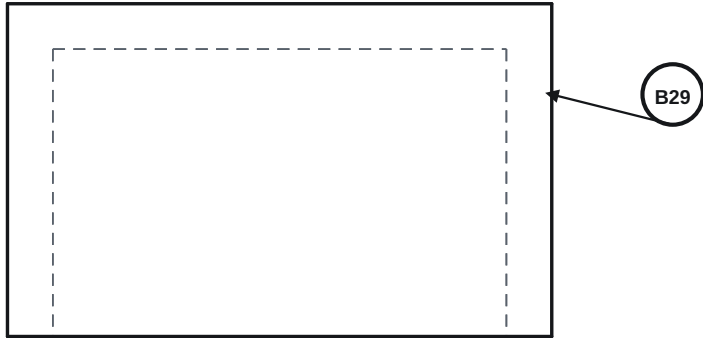


BINDER HOUSING B29

SCALE 2:1 | OCCT HLR



B29 HOUSING 40x36x22



B50 KEYED HALF PROXY

SCALE 5:1 | OCCT HLR

UNVERIFIED series / pinout / SI launch



B50 18x28x4 KEY OFFSET 4 TARGET

B34 FPC CLAMP (LOCAL CSI)

SCALE 5:1 | OCCT HLR



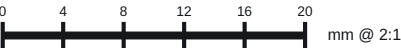
LOCAL CSI KEEPOUT 12 UNVERIFIED; MIPI STAYS IN BINDER

MECHANISM NOTE - BIND-LATCH

States: BLOCKED until pinch $q \geq 2.2$; then $r = 1/2/3.5$ TARGET.
Binder housing moves; B28 shoe stays on carriage. Load path bypasses B50.
First r opens S2 envelope while B50 stays keyed/seated (BRANCH_SAFE dwell).
Release pin retracts only at PINCH_SAFE. Firmware cannot move that pin.
Click-latch section has no sourced tongue STEP; deflection 1.2 TARGET stays a note.

FAILURE / CLEARANCE

Partial click: S2 stays open; Q1 remains off. Key blocks 180 reverse mate.
B50 series/pinout/SI launch UNVERIFIED (#23). Do not invent a connector.
Camera keep-out 32 is not a sourced module outline.
PULL-OFF ≥ 40 N TARGET. Side play ≤ 0.20 TARGET.

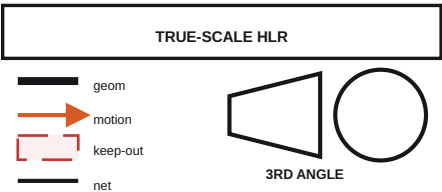


PARTICLE-BASE MANTIS TERRARIUM

UNIVERSAL LATCH + BINDER

DRAFT CAD - NOT SHOP-RELEASE (#31) / NOT FIRST ARTICLE

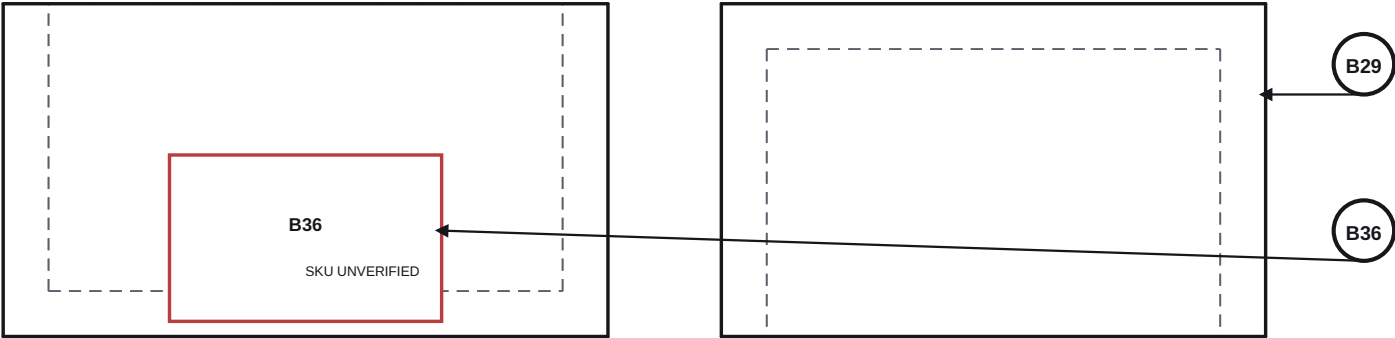
Units: mm | Projection: THIRD-ANGLE (ASME Y14.3) | Color: nets / motion / keep-out only



SCALE 2:1 / 5:1	DATE 2026-08-20	REV B-41P
SHEET 07 OF 12 / S06	STATUS DRAFT CAD	DRAWN GBG#41
DWG MT-S06	FORMAT A3 SVG/PDF	TOL AS NOTED

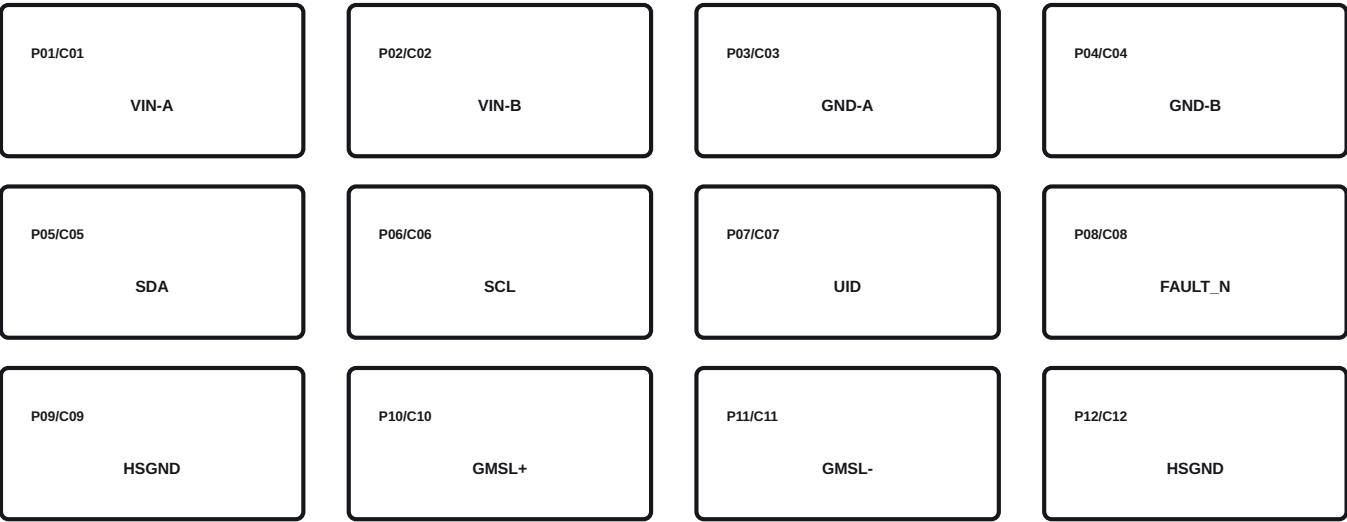
B29 BINDER HOUSING (NO CAMERA SKU STEP)

SCALE 2:1 | OCCT HLR



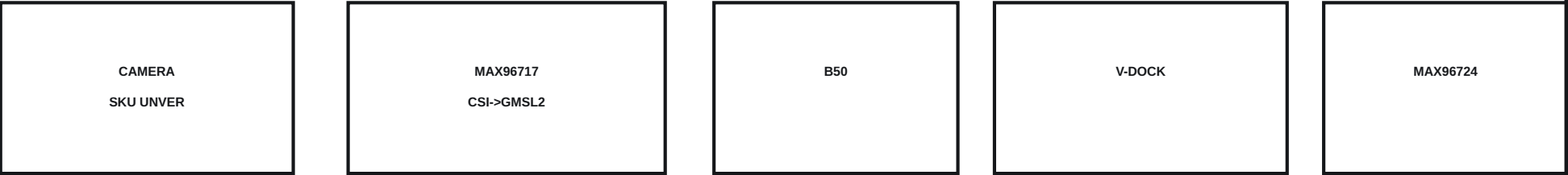
RAIL PXX / BINDER CXX MIRRORED NETS

SCALE NTS - DO NOT SCALE



B50 HLR + VIDEO PATH DIAGRAM

SCALE 5:1 / DIAGRAM | OCCT HLR



RAW MIPI STAYS INSIDE BINDER; GMSL2 CROSSES LOCKED POGO DOCK

MECHANISM NOTE - CAMERA VIDEO DOCK

States: free / S1+S2 power-mated / training / link-trained / pinch-safe.
What moves: binder. V-dock and MAX96724 stay on rail/brick. No tether.
Raw MIPI never rides B50, B27, or the continuous strip (EE-23/25 lock).

FAILURE / WHY / EE-25

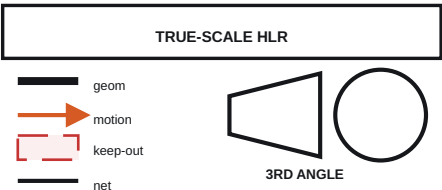
Partial dock or bent HSD: training timeout; Q1 latches off.
Particle lists IMX519 AF and S5K3P9SX sensors; no orderable SKU drawn.
EE-26 coupons: 2x-thru, B27-only, B50-only, full channel. Not animal-use.

PARTICLE-BASE MANTIS TERRARIUM

CAMERA LOAD / SERDES BINDER

DRAFT CAD - NOT SHOP-RELEASE (#31) / NOT FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3) | Color: nets / motion / keep-out only



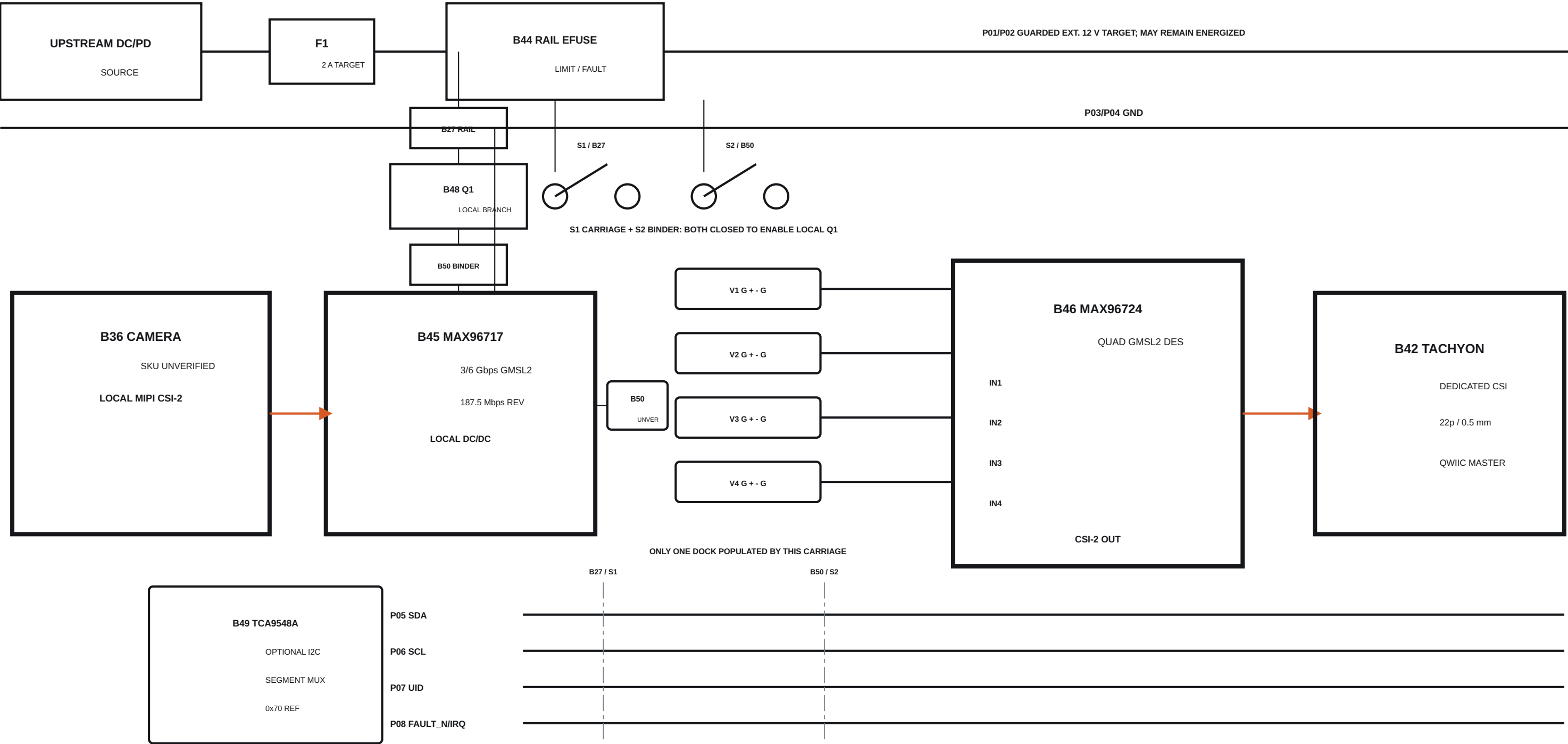
SCALE 2:1 / DIAGRAM	DATE 2026-08-20	REV B-41P
SHEET 08 OF 12 / S07	STATUS DRAFT CAD	DRAWN GBG#41
DWG MT-S07	FORMAT A3 SVG/PDF	TOL AS NOTED

RAIL POWER / CONTROL / GMSL2 - FUNCTIONAL DIAGRAM, NOT RELEASE SCHEMATIC

No STEP projection on this sheet. KiCad on #24/#25 is circuit authority.

COLOR KEY (SHOP LINE ART)

geometry motion keep-out electrical net



MECHANISM NOTE - EE-24 INTERLOCK

EE-24 nodes: VIN_RAIL -> F1 -> VIN_FUSED -> P01||P02 VIN_SHARED.
INTERLOCK_OK = S1 NO series S2 NO. Q1_EN never driven by P08.
S1 open: Q1 off, discharge, ISO_EN off, then B27 may lift.

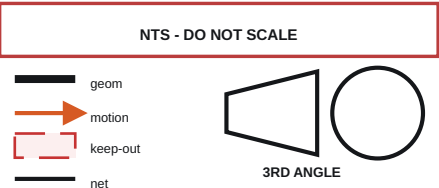
FAULT POLICY / NOT A RELEASE SCHEMATIC

KiCad on #24/#25 is circuit authority; this SVG is a review diagram.
S1/S2/Q1/B27/B50 series UNVERIFIED (#23). No MPNs invented.
P08 diagnostic only. GMSL P10/P11 are not MIPI. EE-26: not animal-use.

PARTICLE-BASE MANTIS TERRARIUM
ELECTRICAL + VIDEO FUNCTIONAL DIAGRAM

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3) | Color: nets / motion / keep-out only



SCALE NTS	DATE 2026-08-20	REV B-41P
SHEET 09 OF 12 / S08	STATUS THEORETICAL	DRAWN GBG#41
DWG MT-S08	FORMAT A3 SVG/PDF	TOL AS NOTED

TACHYON + DESERIALIZER CARRIER

SCALE 1:1 DATASHEET ENVELOPE

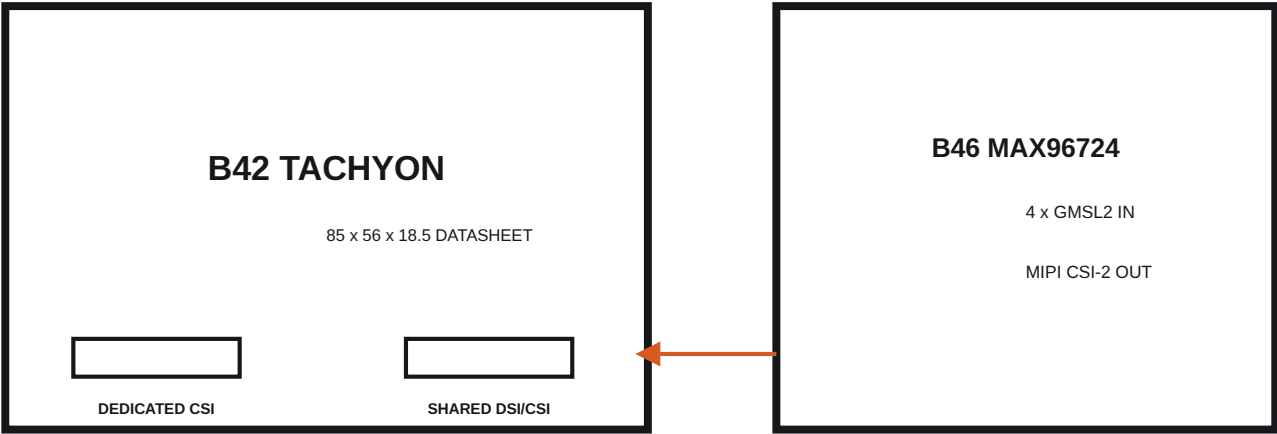
Not a sourced STEP. Envelope from Particle datasheet this run.

OPTIONAL M1 EXTERNAL BRICK

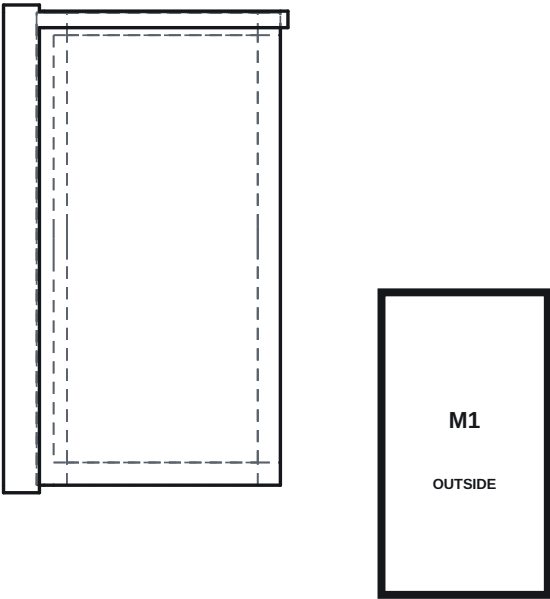
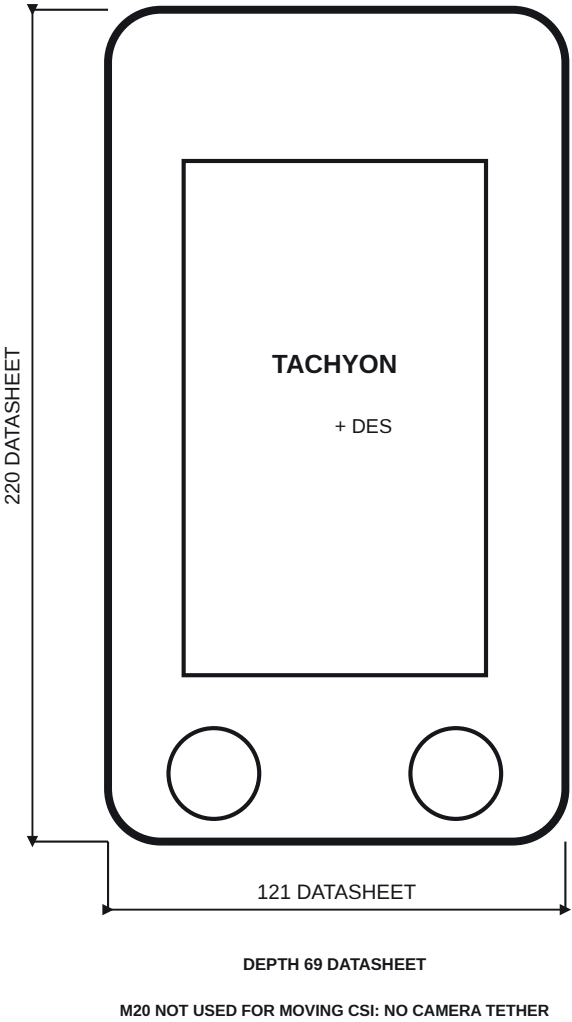
SCALE 1:2 DATASHEET ENVELOPE

EXTERNAL MOUNT VS PR 34 FRONT

SCALE 1:8 | OCCT HLR



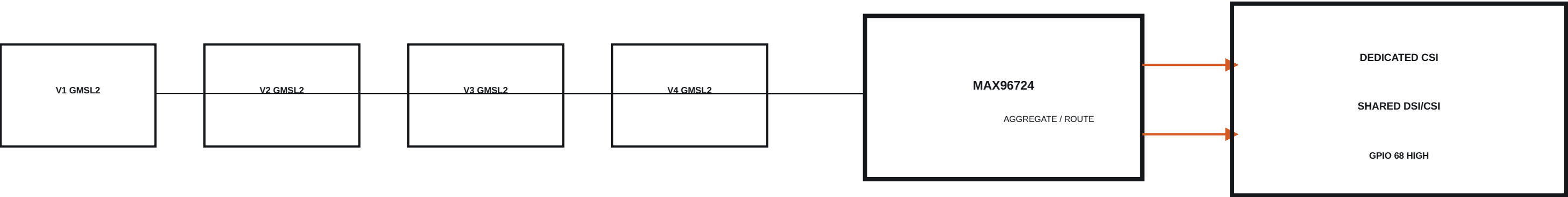
DESERIALIZER CARRIER DIMS / TACHYON DRIVER SUPPORT UNVERIFIED



BRICK + SERDES HUB NEVER ENTER WET VOLUME

FOUR VIDEO DOCKS / TWO TACHYON CSI OPTIONS

SCALE NTS - DO NOT SCALE



VERIFIED THIS RUN (PARTICLE / ADI PAGES)

Tachyon 85x56x18.5; CSI 22p / 0.5 mm; 2.5 Gbps/lane.
GPIO 68 HIGH selects CSI on shared DSI/CSI2. No CSI pin table reprinted.

NOT DRAWN / UNVERIFIED

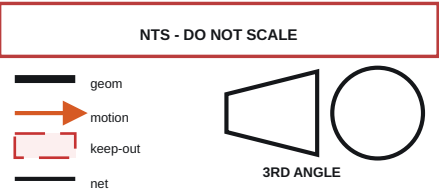
No HAT pin numbers. EE-23 transcribed CSI photos stay in the library.
IMX519/S5K3P9SX are listed sensors; orderable module SKU is not selected.

PARTICLE-BASE MANTIS TERRARIUM

SERDES / CSI / PARTICLE BRICK

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

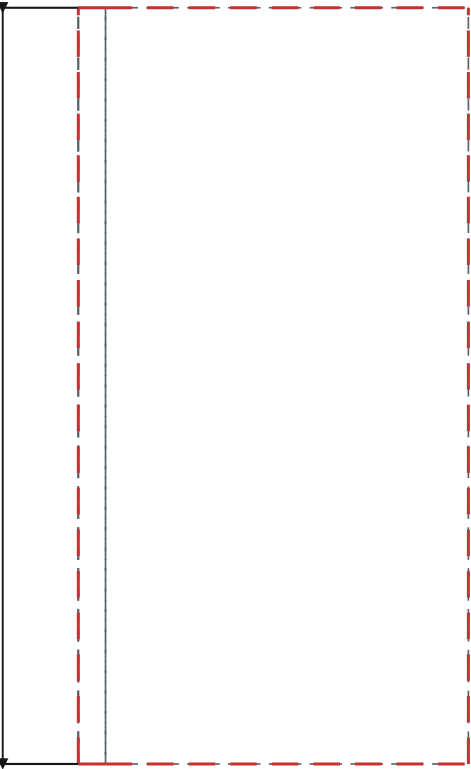
Units: mm | Projection: THIRD-ANGLE (ASME Y14.3) | Color: nets / motion / keep-out only



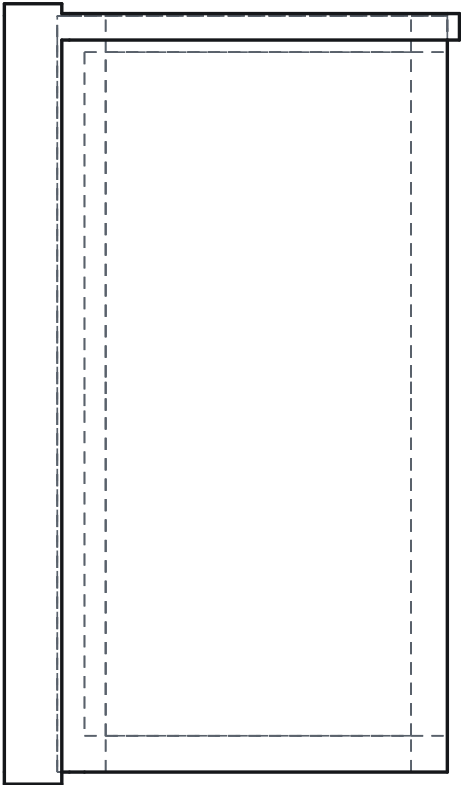
SCALE NTS	DATE 2026-08-20	REV B-41P
SHEET 10 OF 12 / S09	STATUS THEORETICAL	DRAWN GBG#41
DWG MT-S09	FORMAT A3 SVG/PDF	TOL AS NOTED

B20 WET-SIDE BARRIER (KEEP-OUT)

SCALE 1:5 | OCCT HLR
PR 34 B20 STEP fe8f875a80b3 NO METAL IN WET VOLUME



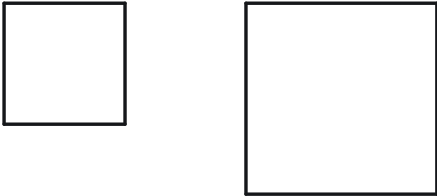
B20 KEEP-OUT



ASSEMBLY HLR

LOW INTAKE B12 / FALSE BOTTOM B14

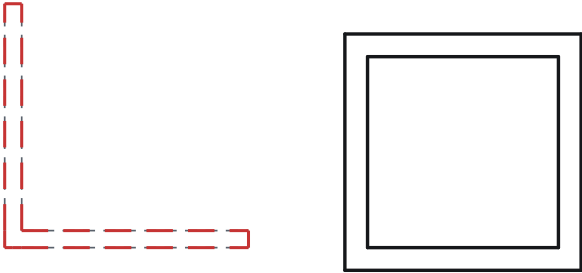
SCALE 1:5 / 1:8 | OCCT HLR



FALSE BOTTOM + 25 TRAY REF

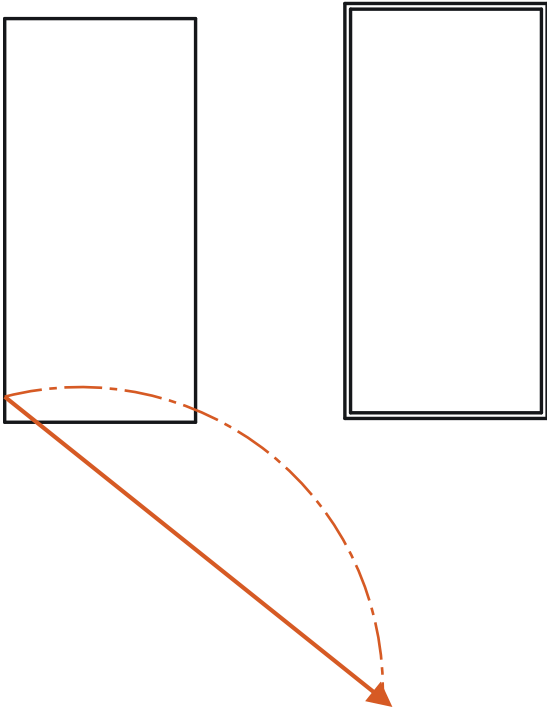
TOP: B20 VS B10 FRAME

SCALE 1:8 | OCCT HLR



DOOR B06 / LABYRINTH B07

SCALE 1:8 | OCCT HLR



90 deg MIN SERVICE SWING (MOTION OVERLAY)

B20 KEEP-OUT (CAD-01 STEP)

Barrier bbox -8..250 X,
-8..250 Y, 0..500 Z.
HLR of B20-animal-wet-barrier.step.
Not a film over contacts.
Molt keep-out 142 CALCULATED.

WET-VOLUME RULE

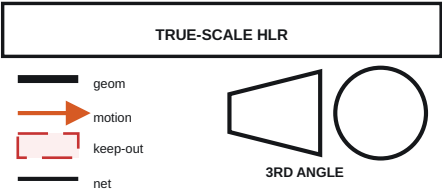
No metal mesh / copper / pogo.
PETG or ASA wet parts.
Nymph gap <=0.50 TARGET.
Solo animal. No taxon inferred.
Heat-mat pass only; no mains.

0 1020304050
mm @ 1:5

PARTICLE-BASE MANTIS TERRARIUM
HUSBANDRY INTERFERENCE

DRAFT CAD - NOT SHOP-RELEASE (#31) / NOT FIRST ARTICLE

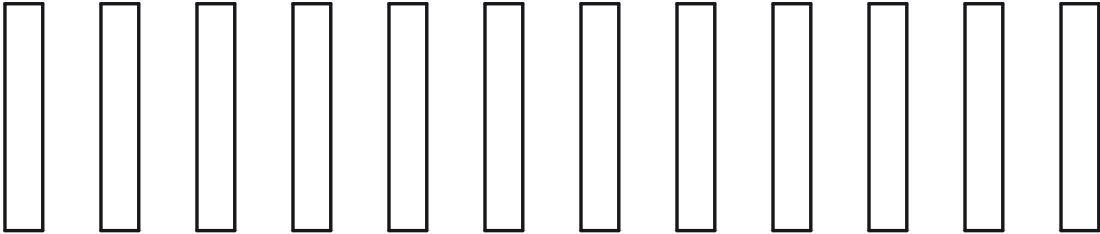
Units: mm | Projection: THIRD-ANGLE (ASME Y14.3) | Color: nets / motion / keep-out only



SCALE 1:5	DATE 2026-08-20	REV B-41P
SHEET 11 OF 12 / S10	STATUS DRAFT CAD	DRAWN GBG#41
DWG MT-S10	FORMAT A3 SVG/PDF	TOL AS NOTED

K - B27 POGO ARRAY PROXY

SCALE 5:1 | OCCT HLR
UNVERIFIED series



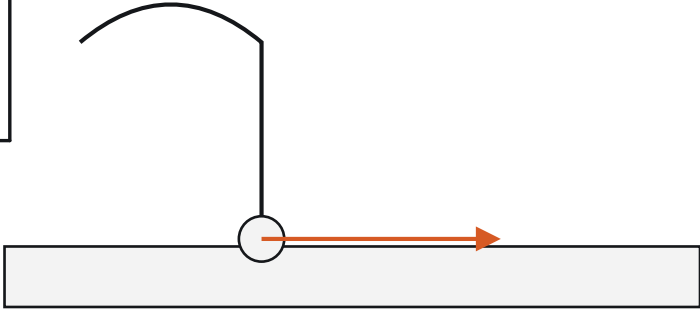
0.60 WORK / 1.20 LIFT TARGET

THROW 2.2 / NEED 1.8

POGO SERIES UNVERIFIED #23

L - STRIP WIPE

SCALE NTS - DO NOT SCALE
No B19 STEP — diagram only

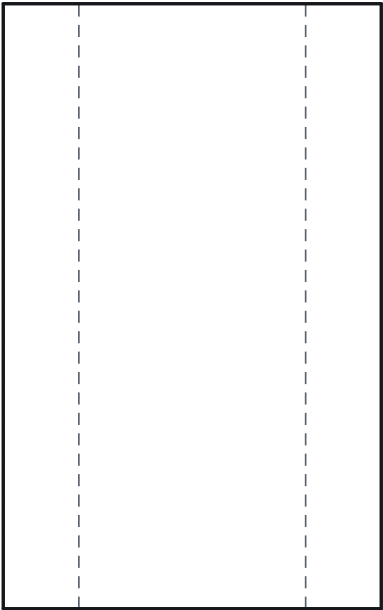


P01-P08 WIPE 0.8 TARGET

P09-P12 VERTICAL SEAT; NO WIPE

M - B25 SPRING POCKET

SCALE 5:1 | OCCT HLR



POCKET +0.30

NO SHARP ROOT

10k CYCLE TARGET

N - B34 CSI CLAMP

SCALE 5:1 | OCCT HLR
Camera FPC length UNVERIFIED

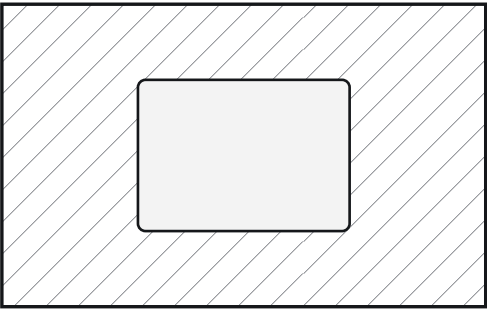


CLAMP FPC INSULATION, NEVER CONDUCTORS

CSI KO 12 UNVERIFIED

P - MAGNET POCKET

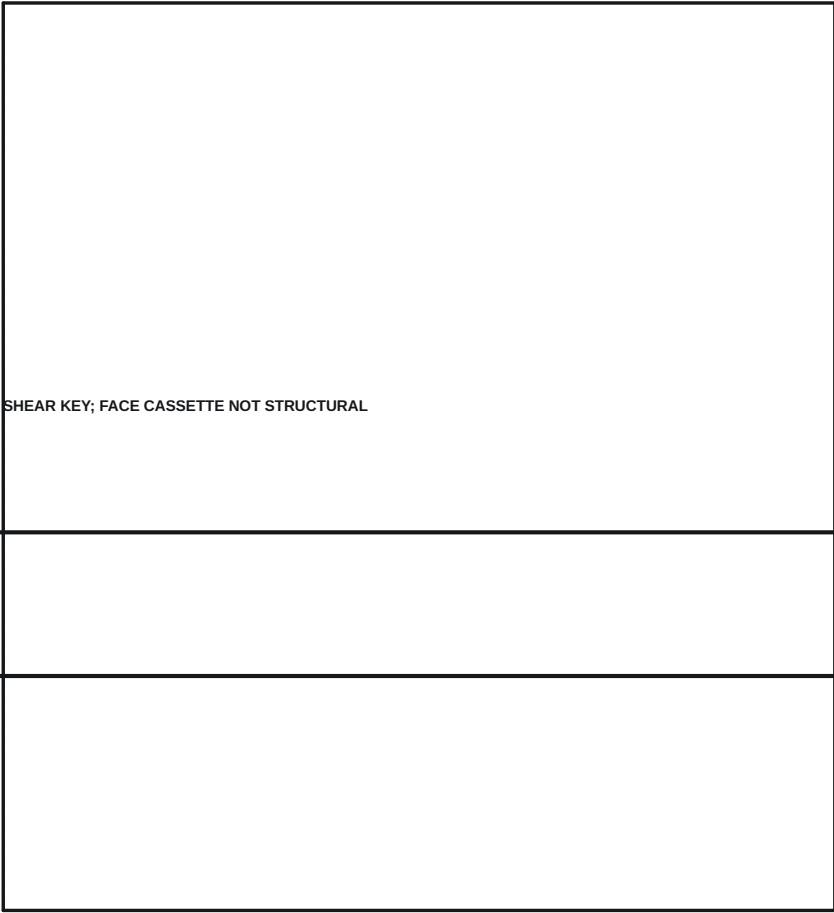
SCALE NTS - DO NOT SCALE
No magnet STEP — UNVERIFIED



CAPTURE LIP; MAGNET DOES NOT DEFINE NYMPH SEAL

Q - B03 250/500 JOINT

SCALE 5:1 | OCCT HLR



SHEAR KEY; FACE CASSETTE NOT STRUCTURAL

DETAIL RELEASE RULE

Projected callouts are HLR of unique-part STEP. TARGET/REF/UNVERIFIED still need coupon or vendor drawing before #31.
B20 keep-out lives on S10. Base CAD-02 cdd523c55a63. CAD-01 fe8f875a80b3. EE-23..26 PRs 37-40 read-only.
Wipe, magnet, and camera-module details remain diagrams: no sourced STEP this run.

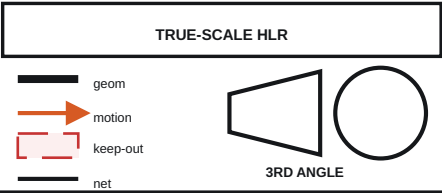


PARTICLE-BASE MANTIS TERRARIUM

DETAIL BLOW-UPS

DRAFT CAD - NOT SHOP-RELEASE (#31) / NOT FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3) | Color: nets / motion / keep-out only



SCALE 5:1 / DIAGRAM	DATE 2026-08-20	REV B-41P
SHEET 12 OF 12 / S11	STATUS DRAFT CAD	DRAWN GBG#41
DWG MT-S11	FORMAT A3 SVG/PDF	TOL AS NOTED